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From Hierarchy to Intelligence

**Author** Jack Dorsey and Roelof Botha**Tags** [Technology](#), [AI](#), [Purpose](#), [Block](#)**Author** Jack Dorsey and Roelof Botha**Tags** [Technology](#), [AI](#), [Purpose](#), [Block](#)

At Sequoia, we see that speed is the best predictor of start-up success. Most companies are focused on AI as a productivity enhancer. Few are focused on the potential of AI to change how we work together. Block is showing what it looks like to fundamentally rethink organization design, ultimately harnessing AI to increase speed as a compounding competitive advantage.

Two thousand years before the first corporate org chart, the Roman Army solved a problem that every large organization still faces: how do you coordinate thousands of people across vast distances with limited communication?

Their answer was a nested hierarchy with a consistent span of control at every level. The smallest unit was the contubernium, eight soldiers who shared a tent, equipment, and a mule, led by a decanus. Ten contubernia formed a century of eighty men under a centurion. Six centuries made a cohort. Ten cohorts made a legion of roughly 5,000. At each layer, a named commander held defined authority, aggregated information from below, and relayed decisions from above. The structure (8 → 80 → 480 → 5,000) was an information routing protocol built around a simple human limitation: a leader can effectively manage somewhere between three and eight people. The Romans discovered this through centuries of warfare. Even today, the US Army's hierarchical chain follows a similar pattern. We now call it "span of control," and it remains the governing constraint of every large organization on earth.

The next big change came from Prussia. After Napoleon's army destroyed the Prussian forces at the Battle of Jena in 1806, a group of reformers led by Scharnhorst and Gneisenau rebuilt the military around an uncomfortable truth: you cannot depend on individual genius at the top. You need a system. They created the General Staff, a dedicated class of trained officers whose job was not to fight but to plan operations, process information, and coordinate across units. Scharnhorst intended these staff officers to "support incompetent Generals,

providing the talents that might otherwise be wanting among leaders and commanders." This was middle management before the term existed. Professionals whose purpose was to route information, pre-compute decisions, and maintain alignment across a complex organization. The military also formalized the distinction between "line" and "staff" functions. Line advances the core mission. Staff provides specialized support. Every corporation still uses this vocabulary today.

Military hierarchy entered the business world through the American railroads in the 1840s and 1850s. The U.S. Army lent West Point-trained engineers to private railroad companies, and these officers brought military organizational thinking with them. Staff and line hierarchies, divisional structure, bureaucratic systems of reporting and control: all of it was developed in the military before the railroads adopted it. In the mid-1850s, Daniel McCallum of the New York and Erie Railroad created the world's first organizational chart to manage a system stretching over 500 miles with thousands of workers. The informal management styles that worked for smaller railroads were failing. Train collisions were killing people. McCallum's chart formalized the same hierarchical logic the Romans had used: layers of authority, defined reporting lines, structured information flow. It became the blueprint for the modern corporation.

Frederick Taylor (1856-1915), often called the "Father of Scientific Management," optimized what happened within that hierarchy. Taylor broke work into specialized tasks, assigned them to trained experts, and managed through measurement rather than intuition. This produced the functional pyramid organization - a structure optimized for efficiency within the information routing system that the military had pioneered and the railroads had commercialized.

The first real stress test of functional hierarchy came during World War II. The Manhattan Project required physicists, chemists, engineers, metallurgists, and military officers to work across disciplinary boundaries toward a single objective under extreme secrecy and time pressure. Robert Oppenheimer organized Los Alamos into functional divisions but insisted on open collaboration across them, resisting the military's instinct to compartmentalize. When the implosion problem became critical in 1944, he reorganized the lab around it, creating cross-functional teams unlike anything in corporate America at the time. It worked, but it was a wartime exception led by a singular figure. The question the postwar business world faced was whether that kind of cross-functional coordination could be made routine.

With the growth and globalization of companies after World War II, the scale limitations of functional design became acute. In 1959, McKinsey's Gilbert Clee and Alfred di Scipio published "Creating a World Enterprise" in the Harvard Business Review, providing an

intellectual framework for a matrix organization that combined functional specialties with divisional units. Under the leadership of Marvin Bower, McKinsey helped companies like Shell and GE implement these principles, balancing central standards with local agility. This became the "professional" or "modern" corporation that propelled the postwar global economy.

Over time, other frameworks emerged to address the complexity, rigidity, and bureaucracy of matrix structures. The McKinsey 7-S framework, developed in the late 1970s by Tom Peters and Robert Waterman, distinguished the "hard Ss" (Strategy, Structure, Systems) from the "soft Ss" (Shared Values, Skills, Staff, Style). The core idea was that structural elements alone were insufficient. Organizational effectiveness required alignment across cultural traits and the human factors that determine whether a strategy actually succeeds. In more recent decades, technology companies have experimented aggressively with organization structure. Spotify popularized cross-functional squads with short sprint cycles. Zappos attempted Holacracy, eliminating management titles entirely. Valve operated with a flat structure and no formal hierarchy. Each of these experiments revealed something about the limitations of traditional hierarchy, but none solved the underlying problem. Spotify moved back toward conventional management as it scaled. Zappos saw significant attrition. Valve's model proved difficult to scale beyond a few hundred people. As organizations grow into the thousands, they revert to hierarchical coordination because no alternative information routing mechanism has been powerful enough to replace it.

The constraint is the same one the Romans faced and the Marine Corps rediscovered in World War II: narrowing span of control means adding layers of command, but more layers mean slower information flow. Two thousand years of organizational innovation has been an attempt to work around this tradeoff without breaking it.

So what's different now?

At Block, we're questioning the underlying assumption: that organizations have to be hierarchically organized with humans as the coordination mechanism. Instead, we intend to replace what the hierarchy does. Most companies using AI today are giving everyone a copilot, which makes the existing structure work slightly better without changing it. We're after something different: a company built as an intelligence (or mini-AGI).

We are not the first to try to move beyond traditional hierarchy. Haier's rendanheyi model, platform organizations, "data-driven" management: these are real attempts at the same problem. What they lacked was a technology capable of actually performing the coordination functions that hierarchy exists to provide. AI is that technology. For the first time, a system can maintain a continuously updated model of an entire business and use it to coordinate work in ways that previously required humans relaying information through layers of management.

For this to work, a company needs two things: a kind of "world model" of its own operations, and a customer signal rich enough to make that model useful.

Block is remote-first. Everything we do creates artifacts. Decisions, discussions, code, designs, plans, problems, and progress all exist as recorded actions. It's the raw material for a company world model. In a traditional company, a manager's job is to know what's happening across their team and relay that context up and down the chain. In a remote-first company where work is already machine-readable, AI can build and maintain that picture continuously. What's being built, what's blocked, where resources are allocated, what's working and what isn't. That's the information the hierarchy used to carry. The company world model carries it instead.

But the capability of the system is only as good as the quality of the customer signal feeding it. And money is the most honest signal in the world.

People lie on surveys. They ignore ads. They abandon carts. But when they spend, save, send, borrow, or repay, that's the truth. Every transaction is a fact about someone's life. Block sees both sides of millions of these transactions every day, the buyer through Cash App

and the seller through Square, plus the operational data from running the merchant's business. That gives the customer world model something rare: a per-customer, per-merchant understanding of financial reality built from honest signal that compounds. The richer the signal, the better the model. The better the model, the more transactions. The more transactions, the richer the signal. Together, the company world model and the customer world model form the foundation for a different kind of company. Instead of product teams building predetermined roadmaps, you build four things.

First, capabilities. The atomic financial primitives: payments, lending, card issuance, banking, buy-now-pay-later, payroll, and so on. These are not products. They are building blocks that are hard to acquire and maintain (some have network effects and regulatory permission). They have no UIs of their own. They have reliability, compliance, and performance targets.

Second, a world model. This has two sides. The company world model is how the company understands itself and its own operations, performance, and priorities, replacing the information that used to flow through layers of management. The customer world model is the per-customer, per-merchant, per-market representation built from proprietary transaction data. It starts with raw transaction data today and evolves toward full causal and predictive models over time.

Third, an intelligence layer. This is what composes capabilities into solutions for specific customers at specific moments and delivers them proactively. A restaurant's cash flow is tightening ahead of a seasonal dip the model has seen before. The intelligence layer composes a short-term loan from the lending capability, adjusts the repayment schedule using the payments capability, and surfaces it to the merchant before they even think to look for financing. A Cash App user's spending pattern shifts in a way the model associates with a move to a new city. The intelligence layer composes a new direct deposit setup, a Cash App Card with boosted categories for their new neighborhood, and a savings goal calibrated to their updated income. No product manager decided to build either solution. The capabilities existed. The intelligence layer recognized the moment and composed them.

Fourth, interfaces (hardware and software). Square, Cash App, Afterpay, TIDAL, bitkey, proto. These are delivery surfaces through which the intelligence layer delivers composed solutions. They are important, but they are not where the value is created. The value is in the model and the intelligence.

When the intelligence layer tries to compose a solution and can't because the capability doesn't exist, that failure signal is the future roadmap. The traditional roadmap, where product managers hypothesize about what to build next, is any company's ultimate limiting factor. In this model, customer reality generates the backlog directly.

If this is what the company builds, then the question becomes: what do the people do?

The org structure follows from this, and it inverts the traditional picture. In a conventional company, the intelligence is spread throughout the people and the hierarchy routes it. In this model, the intelligence lives in the system. The people are on the edge. The edge is where the action is.

The edge is where the intelligence makes contact with reality. People reach into places the model can't go yet. They sense things the model can't perceive: intuition, opinionated direction, cultural context, trust dynamics, the feeling in a room. They make the calls the model shouldn't make on its own, especially ethical decisions, novel situations, and high-stakes moments where the cost of being wrong is existential. A world model that can't touch the world is just a database. But the edge doesn't need layers of management to coordinate it. The world model gives every person at the edge the context they need to act without waiting for information to travel up and down a chain of command.

In practice, this means we normalize down to three roles.

Individual contributors (ICs) who build and operate capabilities, the model, the intelligence layer, and the interfaces. They are deep specialists and experts in a specific layer of the system. The world model provides the context that a manager used to provide, so ICs can make decisions about their layer without waiting to be told what to do.

Directly Responsible Individuals (DRI) who own specific cross-cutting problems or opportunities and customer outcomes. A DRI might own the problem of merchant churn in a specific segment for 90 days, with full authority to pull resources from the world model team, the lending capability team, and the interface team as needed. DRIs may persist on certain problems or move elsewhere to solve new ones.

Player-coaches who combine building with developing people. They replace the traditional manager whose primary job was information routing. A player-coach still writes code or builds models or designs interfaces. They also invest in the growth of the people around them. They don't spend their days in status meetings, alignment sessions, and priority negotiations. The world model handles alignment. The DRI structure handles strategy and priority. The player-coach handles craft and people.

There is no need for a permanent middle management layer. Everything else the old hierarchy did, the system coordinates, and everyone is empowered, with a role that's much closer to the work and the customer.

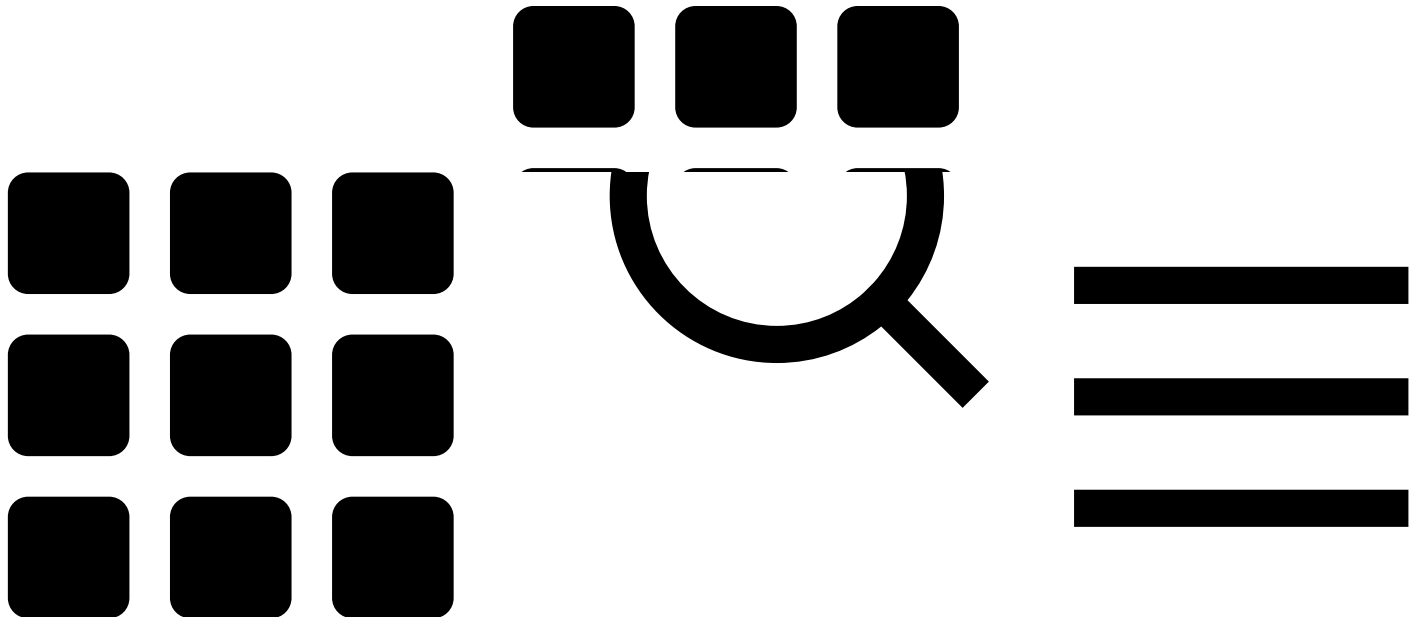
Block is in the early stages of this transition. It will be a difficult one, and parts of it will likely break before they work. We're writing about it now because we believe every company will eventually need to confront the same question we did: what does your company understand that is genuinely hard to understand, and is that understanding getting deeper every day?

If the answer is nothing, AI is just a cost optimization story. You cut headcount, improve margins for a few quarters, and eventually get absorbed by something smarter. If the answer is deep, AI doesn't augment your company. It reveals what your company actually is.

Block's answer is the economic graph: millions of merchants and consumers, both sides of every transaction, financial behavior observed in real time. That understanding compounds every second the system operates. We believe the pattern behind this, a company organized as an intelligence rather than a hierarchy, is significant enough that it will reshape how companies of all kinds operate over the coming years. Block is far enough along to show the idea is more than theory (though, we welcome debate and feedback to pressure test and improve our ideas).

Companies move fast or slow based on information flow. Hierarchy and middle management impede information flow. For two thousand

years, from the Roman contubernium to today's global enterprises, we have had no real alternative. Eight soldiers sharing a tent needed a decanus. Eighty men needed a centurion. Five thousand needed a legate. The question was never whether you needed layers. The question was whether humans were the only option for what those layers do. They aren't anymore. Block is building what comes next.



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